

Total No. of Questions : 6]

SEAT No. :

P69

[Total No. of Pages : 2

APR - 18/BE/Insem. - 27

B.E. (Mechanical Engineering)

DESIGN OF PUMPS, BLOWERS AND COMPRESSORS

(2012 Course) (Elective - IV) (Semester - II) (402050 C)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain the basic equation of Energy Transfer between fluid and rotor. [6]

b) Define the following terms. [4]

- i) Compressible Flow Machines
- ii) Incompressible Flow Machines
- iii) Pumps
- iv) Turbines

OR

Q2) a) Write a short note on "Velocity Triangles" [5]

b) Write a short note on "Dimensionless Parameters" [5]

Q3) a) Explain Air Vessel in Reciprocating Pumps? [5]

b) A single acting reciprocating pump running at 50 r.p.m. delivers $0.01 \text{ m}^3/\text{s}$ of water. The diameter of the piston is 200 mm and stroke length 400mm determine [5]

- i) The theoretical discharge of the pump
- ii) Co-efficient of discharge and
- iii) Slip and the percentage slip of the pump.

OR

P.T.O.

Q4) a) Derive the expression for discharge, work done and power required to drive a double acting reciprocating pump with neat sketch? [5]

b) A double acting reciprocating pump running at 40 r.p.m. is discharging 1.0 m^3 of water per minute. The pump has a stroke of 400 mm. The diameter of the piston is 200 mm. The delivery and suction head are 20 m and 5 m respectively. Find the slip of the pump and power required to drive the pump? [5]

Q5) a) Explain the basic design criteria of pump. [5]

b) Explain various forms of corrosion occurred in hydraulic machines. [5]

OR

Q6) a) Explain in detail different types of Impellers. [5]

b) Explain the hydraulic design of pumps in details. [5]

